Muhammad Ahmed

+91 9353486865 ahmedmu7077@gmail.com linkedin.com/in/muhammad-ahmed github.com/ahmed7077
portfolio

Experience

LLM Intern | **IonIdea** July 2025 - September 2025

Github: https://github.com/ahmed7077/efficient-llm-adaptation

- Fine-tuned Meta Llama 3.2-3B-Instruct on 430 instruction-response pairs using LoRA with Hugging Face Transformers, TRL, and PEFT, completing 800 training steps (~8 epochs) in ~27 minutes on an NVIDIA T4 GPU.
- Reduced training loss from 1.35 to 0.51 while adapting the model for domain-specific instruction following using LoRA (r=16, α=32).
- Integrated a Model Context Protocol (MCP)-based retrieval layer for structured knowledge-base lookups with fallback to the fine-tuned LLM, improving domain-specific response reliability.

Projects

Facial Expression Recognition System | *Python, OpenCV, CNN, NumPy*

- Built a real-time facial emotion recognition system using EfficientNetB0, trained on the FER2013 dataset across seven emotion classes.
- Implemented an OpenCV inference pipeline for face detection, preprocessing, and real-time emotion classification.
- Evaluated model performance across anger, disgust, fear, happiness, sadness, surprise, and neutral expressions.

Astronomical Event Detection System | *Python, Flask, OpenCV, NumPy, Raspberry Pi*

- Developed a Raspberry Pi-based computer-vision system to detect and classify meteors, satellites, and star clusters from captured imagery.
- Applied OpenCV-based preprocessing, thresholding, contour analysis, and object-detection techniques to analyze captured imagery.

IoT-Based Server Monitoring System | *C++, Arduino, Sensors (DHT11, MQ2), ThingSpeak*

- Developed an IoT solution to monitor server room conditions using Arduino and sensor modules.
- Collected and displayed real-time environmental data for preventive maintenance.
- Enabled early detection of abnormal operating conditions through continuous monitoring.

Education

B.Tech in Information Science and Engineering-*Presidency University, Bengaluru, India* | CGPA:7.70	2023 – 2027
Pre-University (PCMCs)-*Whitefield Global School (Deeksha), Bengaluru, India* | 82%	2021 – 2023
10th ICSE-*Ryan International School, Bengaluru, India* | 81%	2008 – 2021

Skills

Programming Languages: Python, Java, C, C++, SQL
AI/ML: Machine Learning, Deep Learning, LLM Fine-Tuning, NLP, Computer Vision, CNNs
LLM Technologies: LoRA, PEFT, Retrieval-Based LLM Systems
Frameworks & Libraries: PyTorch, Hugging Face Transformers, TRL, OpenCV, Keras, scikit-learn, Flask
Web: HTML, CSS, JavaScript, React
Databases: MySQL
Tools & Platforms: Git, GitHub, Google Colab, VS Code, Ollama, Raspberry Pi, Arduino

Certifications

- IBM – Machine Learning with Python | September 2026
- IBM – Introduction to Deep Learning & Neural Networks with Keras | September 2026
- IBM – Artificial intelligence Fundamentals | September 2026
- Harvard University – CS50x: Introduction to Computer Science | August 2026
- UpGrad (Microsoft) – Generative AI Foundations | June 2026